PROJECT DESIGN PHASE

Solution Architecture

Date	14 October 2026
Project Name	Automated Network Request Management in ServiceNow
Maximum Marks	4 Marks

Solution Architecture

The solution architecture connects the network-service request problem with the ServiceNow components used in the current implementation. It defines how requester inputs move from the Service Catalog through variable retrieval, operational-record creation, approval, notification, and request lifecycle tracking, while also showing the responsibilities of approvers, IT administrators, and the network fulfilment team.

• Solve the problem of fragmented and manually coordinated network-service requests through a centralized digital workflow.

• Describe the structure and behaviour of the ServiceNow solution from request submission through approval and fulfilment handoff.

• Define the major functional blocks used in the current implementation and show how request data and control decisions move between them.

• Support maintainability through Flow Designer-first automation, structured operational data, controlled access, and traceable execution.

Solution Architecture Diagram

USER / ACCESS LAYER	SERVICENOW APPLICATION LAYER	APPROVAL / OPERATIONS LAYER
Requester / End User	**Network Request Catalog Item**	**Approver**
ServiceNow Service Portal	Catalog Variables + Requester Information	Approval Request Relationship
	Catalog UI Policy	Network Fulfilment Team
	Flow Designer	**IT Administrator**
	REQ / RITM	**Roles & ACLs**
	Database Tables (u_database_tables)	**Flow Execution Details / Logs**
	Notifications	

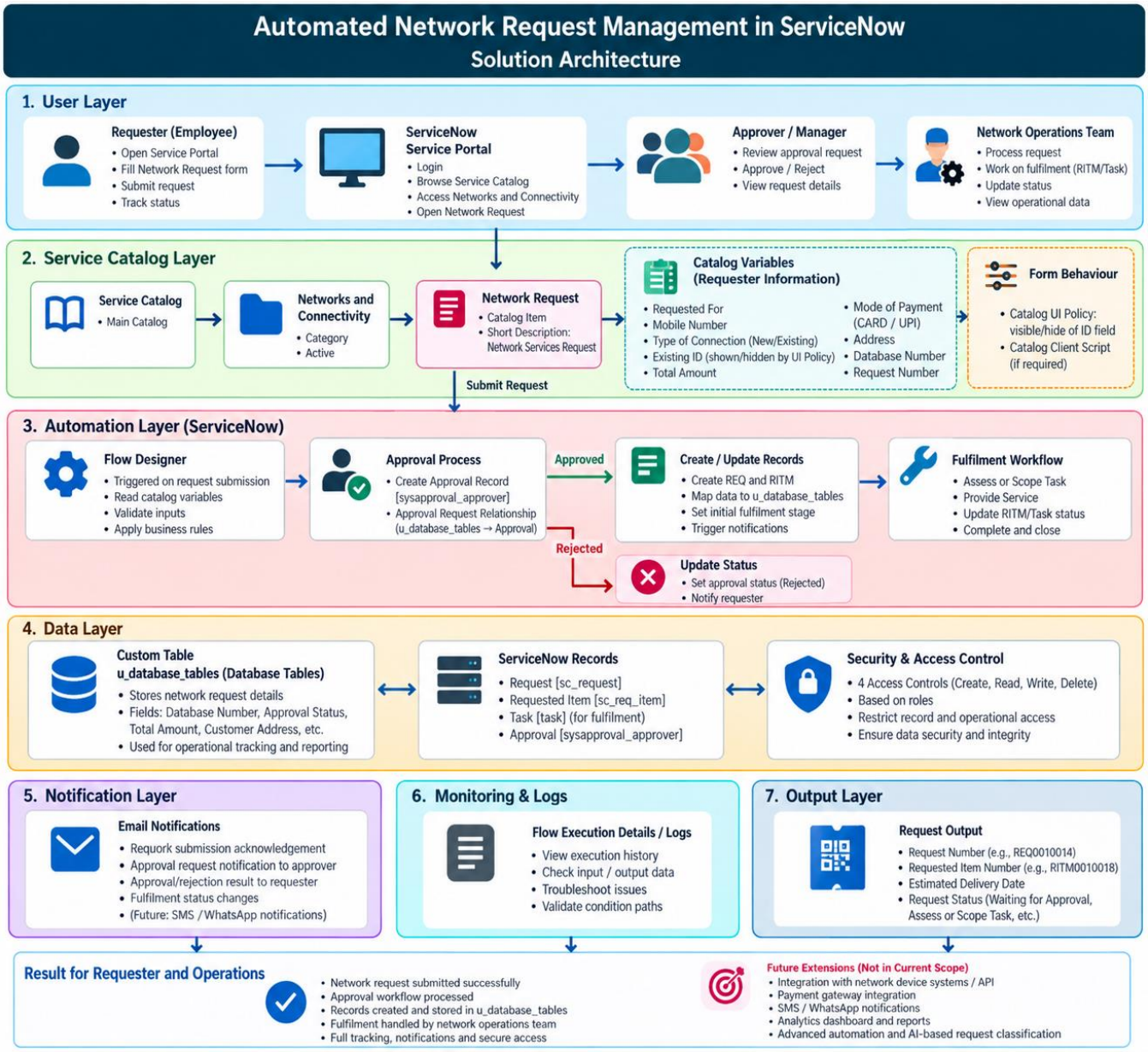


Figure 1. Solution Architecture – Automated Network Request Management in ServiceNow

Architecture Components and Responsibilities

S.No.	Component	Architecture Role	Responsibility
1	**Requester / End User**	Primary request initiator	Submits the Network Request, provides required details, receives confirmation, and tracks the request lifecycle.
2	**ServiceNow Service Portal / Service Catalog**	User access and request-entry layer	Provides the Network Request catalog item under Networks and Connectivity and presents the structured request form.
3	**Network Request Catalog Item**	Structured request capture	Captures the configured request variables such as Requested For, Mobile Number, Type of Connection, Existing ID, Total Amount, Mode of

			Payment, Address, Database Number, and Request Number.
4	**Requester Information Variable Set**	Reusable requester-information grouping	Groups requester-related information used by the catalog form.
5	**Catalog UI Policy**	Dynamic form behaviour	Controls conditional field visibility; the current implementation includes the ID-field visibility behaviour.
6	**Flow Designer**	Primary workflow automation engine	Retrieves catalog variables, creates the operational record, sends notifications, asks for approval, evaluates the outcome, and updates the record.
7	**REQ / RITM**	Native ServiceNow request lifecycle	Maintains the submitted request and requested-item context, including request number, RITM, stage, and state.
8	**Database Tables (u_database_tables)**	Operational data record	Stores mapped request information such as requester, mobile number, connection type, amount, payment mode, address, database number, request number, and approval status.
9	**Approval Request Relationship**	Approval visibility / relationship	Relates Approval records to the corresponding Database Tables record using source table and document ID filtering.
10	**Approver**	Decision point	Reviews the request through the configured approval action; the current flow uses an approval decision to determine the next branch.
11	**Roles & ACLs**	Security and access control	Controls Create, Read, Write, and Delete access for the configured operational table according to ServiceNow security configuration.
12	**Notifications**	Communication layer	Sends configured request-created and request-resolved email notifications from the automated flow.
13	**Flow Execution Details / Logs**	Observability and troubleshooting	Supports investigation of variable mapping, conditions, approval paths, record creation, and notification execution.
14	**Network Fulfilment Team**	Operational fulfilment layer	Receives the approved work context and performs the network-service fulfilment activities that require operational action.

End-to-End Data and Control Flow

1. The requester opens the ServiceNow Service Portal and selects the Network Request catalog item under Networks and Connectivity.

2. The requester enters the required catalog information, including requester details, mobile number, connection type, amount, payment mode, address, database number, and request number as configured.

3. The Catalog UI Policy controls the relevant dynamic form behaviour, including conditional visibility of the ID field.

4. The requester submits the catalog request, creating the native ServiceNow REQ/RITM request context.

5. Flow Designer retrieves the catalog variables from the Network Request submission.

6. The flow creates a record in the Database Tables (u_database_tables) operational table and maps the required request values into that record.

7. The flow sends a request-created notification to communicate that the request has been received.

8. The flow asks for approval and waits for the configured approval outcome.

9. If the approval result is Approved, the flow updates the operational record and sends the request-resolved notification. The request can then proceed through the fulfilment/status lifecycle.

10. If the approval result is not approved, the flow updates the operational record with the rejected outcome and routes the request to the configured alternative path.

11. The requester, approver, IT administrator, and fulfilment team use the appropriate ServiceNow records and status information according to their responsibilities.

12. Administrators can inspect Flow Execution Details and platform logs to troubleshoot mapping, approval, condition, or notification issues.

Architecture Design Principles

Principle	Application in this Project
Low-code first	Flow Designer is the primary automation mechanism; custom scripting is kept limited to configuration needs.
Separation of concerns	Portal/catalog, form behaviour, workflow automation, operational data, approval, security, notification, and troubleshooting are treated as distinct responsibilities.
Centralized request data	Submitted information is mapped into a structured operational record instead of remaining only in the catalog submission.
Controlled access	Roles and ACLs provide controlled access to the operational table and administrative capabilities.
Traceable lifecycle	REQ/RITM context, approval state, operational record status, and notifications provide a traceable request path.
Maintainability	The configuration-first architecture is intended to be understandable and modifiable by ServiceNow administrators.

Current Boundary and Future Expansion

The current architecture covers catalog-based request capture, dynamic form behaviour, Flow Designer automation, REQ/RITM request context, the Database Tables operational record, approval processing, notifications, access control, and workflow troubleshooting. The following items are future expansion points rather than completed capabilities:

• Additional network request categories and specialized catalog items.

• Multi-level or role-specific approval chains beyond the current configured approval path.

• Automated fulfilment through network-device or external infrastructure APIs.

• Advanced SLA, escalation, dashboard, and operational analytics capabilities.

• External SMS, WhatsApp, or other messaging channels.

• AI-assisted request classification, prioritization, anomaly detection, or intelligent routing.

End of Solution Architecture